Most physicists love Einstein's relativistic laws and are loathe to give them up, and so lean toward repulsion. The hypothetical material that repels has been given the name "dark energy."

The final verdict is not in. But if the cause of the anomaly is, indeed, dark energy (whatever that may be), then gravitational observations now tell us that 68 percent of the universe's mass is in dark energy, 27 percent is in dark matter, and only 5 percent is in the kind of ordinary matter of which you, I, planets, stars, and galaxies are made.

So physicists today have another holy grail: to understand whether the universe's accelerated expansion is caused by a breakdown of Einstein's relativistic laws (and if so, what is the nature of the correct laws?), or is caused by repulsive dark energy (and if so, what is the nature of the dark energy?).

Gravitational Anomalies in *Interstellar*



The gravitational anomalies in *Interstellar* are seen on Earth, by contrast with the three anomalies that I described.

Physicists have put great effort into searching for such anomalies on Earth, beginning with Isaac Newton himself in the late 1600s. Those searches have produced many claimed anomalies, but all claims, upon deeper scrutiny, have collapsed.

The anomalies in *Interstellar* are startling for their weirdness and strength, and the way they change as time passes. If anything like them had occurred in the twentieth century or early twenty-first, physicists would surely have noticed them and explored them with great fervor. Somehow, gravity on Earth has been altered in the era of *Interstellar*.

And, indeed, Romilly tells Cooper so in the movie: "We started detecting gravitational anomalies [on Earth] almost fifty years ago," and